

MolInterAct: Multiscale Cross-modal Interaction for Robust Molecular Representation Learning

Mengwei Sun¹, Chengwei Ai¹, Diya Zhang¹, Xiaoyi Liu², Qiaozhen Meng^{3*}, Shiqiang Ma^{4*}, Fei Guo^{1*}

¹*School of Computer Science, Central South University, Changsha, China*

²*School of Chinese Materia Medica, Beijing University of Chinese Medicine, Beijing, China*

³*School of Computer Science, Xiangtan University, Xiangtan, China*

⁴*Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China*

Email: qiaozhenm@xtu.edu.cn, sq.ma@siat.ac.cn, guofei@csu.edu.cn

* Corresponding Author: Qiaozhen Meng, Shiqiang Ma, Fei Guo

Abstract—Molecular representation learning, which captures the fundamental characteristics of chemical compounds, is crucial for AI-driven drug discovery. Existing methods integrate various modalities (e.g., 2D topology and 3D geometry) to develop robust representations. However, current multi-modal fusion strategies either align embedding space through independent models separately, thereby overlooking complementary information, or bridge modalities at a coarse-grained level, failing to capture inherent correlations. We present MolInterAct, an innovative pre-training framework designed to promote multiscale interactions between 2D and 3D modalities at both atomic-level and molecule-level. Specifically, we propose a fine-grained fusion module, coupled with a customized complementary masking strategy, to seamlessly integrate information at the atomic-level, mitigating overlap and similarity between 2D and 3D representations. In addition, we introduce a fusion contrastive module, which operates at the molecule level, to further strengthen the fusion of 2D and 3D representations while preserving modality-specific features. Finally, we incorporate an intra-modal reconstruction module to reconstruct the original information, further refining the model's understanding of individual modality. Extensive experiments demonstrate that our model outperforms existing molecular pretraining methods across both 2D and 3D benchmarks, highlighting the effectiveness of multiscale fusion between modalities.

Index Terms—Drug Discovery, Molecular Representation Learning, Molecular Pretraining, Multi-modal Fusion.

I. INTRODUCTION

Robust molecular representations are fundamental for AI-driven drug discovery [1]–[3], enabling models to capture intricate molecular properties and tackle various tasks like drug screening [4], optimization [5] and design [6]. Molecules are represented in diverse modalities, encompassing 1D, 2D and 3D formats. Since different modalities offer complementary perspectives, relying on a single modality often fails to comprehensively capture molecular properties, limiting the model's performance on various tasks.

Current multi-modal learning approaches seek to integrate information from diverse sources to achieve a more comprehensive understanding [7]–[9]. The core challenge lies in effectively interacting and aligning different modalities to model their complex relationships. As illustrated in Figure 1, two predominant strategies address this problem: interaction-based and generation-based methods. Interaction-based ap-

proaches focus on integrating feature representations from different modalities to learn cross-modal relationships, often employing contrastive learning to align embeddings of the same molecule [7]–[9]. In contrast, generation-based methods use one modality as a constraint to bridge to another modality's space, enabling mutual data generation through generative models [8], such as transforming between 2D and 3D representations. While interaction-based methods ensure modal consistency, they often fail to capture complex cross-modal dependencies; generation-based approaches prioritize output quality over direct modal interaction, potentially suboptimal for fine-grained information exchange. Therefore, exploring fine-grained interaction mechanisms between modalities remains crucial for advancement.

To address the aforementioned challenges, we propose MolInterAct (Figure 2), a multi-modal pretraining framework that facilitates multi-scale interactions between 2D topological and 3D geometric representations. Our approach begins with a complementary masking strategy applied to both modalities, where atoms are masked non-overlappingly to encourage cross-modal guidance during reconstruction. The masked inputs are processed by separate modality-specific encoders to extract intrinsic features, followed by a fine-grained fusion module that deeply integrates unmasked token embeddings from both modalities into a unified atomic-level representation. To strengthen global alignment, we introduce a fusion contrastive learning module that aligns the fused representation with individual modality representations at the molecule level. Finally, the unified representations are split and reconstructed via separate decoders. Extensive experiments show that MolInterAct outperforms existing molecular representation methods across a wide range of benchmarks, whether using 2D, 3D, or multi-modal inputs.

Our contributions are summarized as follows:

- We present MolInterAct, an innovative molecular pretraining framework that leverages the complementary information from topology and geometry representations while facilitating multiscale cross-modal interactions, ultimately leading to enhanced performance.
- We propose a simple yet effective fine-grained fusion module and introduce a fusion contrastive learning (FCL)

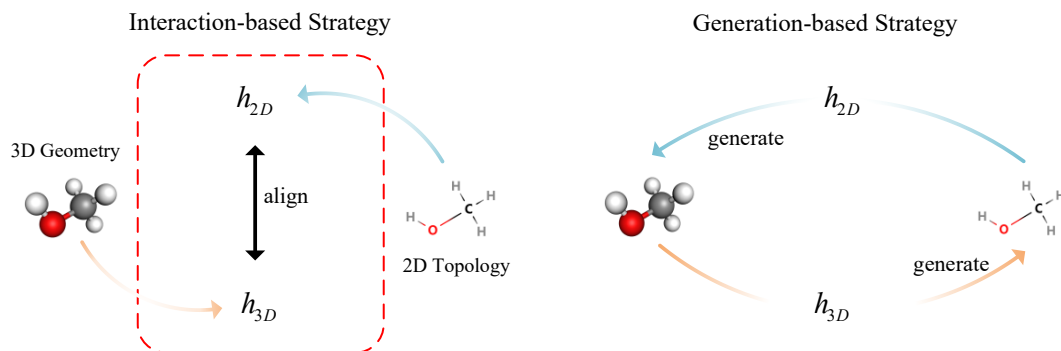


Fig. 1: Illustration of two prevailing multi-modal learning strategies: Interaction-based (left) and Generation-based (right). The former aims to align embedding space of different modalities, which usually adopts contrastive learning method. The latter relies on bridging two modalities via mutual construction.

module, to enable both fine-grained (atomic-level) and coarse-grained (molecule-level) interactions between the modalities.

- Through comprehensive experiments on biological tasks in MoleculeNet and quantum tasks in QM9, our model outperforms existing molecular pretraining methods, demonstrating its superior performance.

II. METHOD

A. Single-Modal Feature Learning

A molecule $\mathcal{M}$ consists of a collection of atoms held together by attractive forces. The structure of $\mathcal{M}$, can be represented in different forms, such as 2D graph structure and 3D geometric structure. For the 2D graph structure, atoms serve as nodes and covalent bonds as edges. We define $\mathcal{M}_{2D} = (\mathbf{X}, \mathbf{E})$, where $\mathbf{e}_{(i,j)} \in \mathbf{E}$ denotes the edge feature (the type of the bond) between atom i and j if the edge exists. For the 3D geometric structure, atoms are treated as point clouds in 3D Euclidean space. We define $\mathcal{M}_{3D} = (\mathbf{X}, \mathbf{R})$, where $\mathbf{R} = \{\mathbf{r}_1, \dots, \mathbf{r}_n\}$, $\mathbf{r}_i \in \mathbb{R}^3$.

In this work, we mask atom features of the 2D topology [10], and the coordinates of atoms in 3D geometry. Since the positions are continuous values, not discrete values, we cannot use a special value to represent the mask [11]. Thus, we add noise to the ground-truth positions to corrupt the 3D positions, and the model is trained to predict noise.

At this stage, we aim to preserve the integrity and distinctive structural characteristics of each modality, ensuring their unique features remain unaltered throughout the processing pipeline. This foundation enables more accurate and effective cross-modal interaction in subsequent phases. We employ two separate encoders to project both modalities into embeddings of the same dimensionality for further integration.

B. Fine-grained Cross-modal Fusion Module

Complementary Masking. Existing multi-modal pretraining methods focus extensively on model architecture and pretraining objective, while there is limited exploration of multi-modal masking strategy [7], [8].

Most approaches use uniform or random masking, where tokens from different modalities are concatenated into a sequence to perform modality fusion [12], [13]. This may result in feature redundancy due to the overlap or similarity between the two modalities, particularly in lower-level representations of molecular structures.

In contrast, our complementary masking strategy selectively obscures different parts of the 2D and 3D information, leveraging the unique strengths of each modality. The 2D topology captures atomic connections and bonds, while the 3D structure provides spatial conformation and geometric properties. By masking complementary parts of the two modalities, the model is forced to learn from one modality using the other, allowing it to integrate the distinct characteristics of both 2D and 3D data and facilitating the integration of complementary information. As a result, the model gains a deeper understanding of the relationship between 2D and 3D data, leading to more robust molecular representations that enhance performance on downstream tasks.

Cross-Modal Fusion. We develop a unified Transformer-based backbone without positional embeddings to perform feature interaction [14], as we expect that the specific encoders to have already encoded the structural information into the node embeddings. By leveraging Transformer’s attention mechanism, the model can effectively capture and learn the relationships between different input token embeddings across two modalities. The Self-Attention mechanism is formulated as:

$$\text{Self-Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}, \quad (1)$$

We remove the masked tokens, add modality embeddings to the remaining and then concatenate them. Due to complementary masking, the concatenated tokens form a complete molecular representation, with atom tokens from both 2D and 3D modalities. This approach ensures that different parts of the 2D and 3D information are obscured, avoiding redundancy or overlap between the representations and enabling a more com-

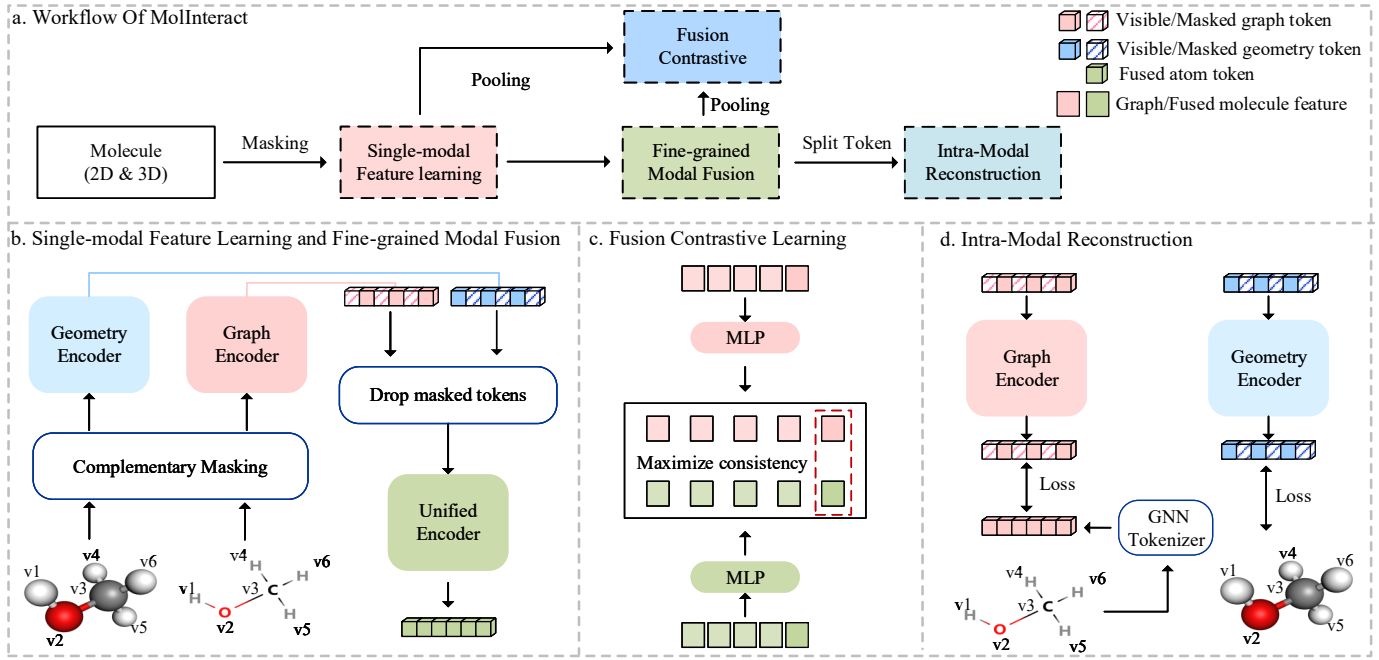


Fig. 2: Pre-training pipeline for MolInterAct. The point cloud and graph are masked in a complementary manner, after which both modalities are processed through their respective encoders. After the embedding stage, tokens undergo the fine-grained fusion module, enabling fusion at atomic level. Next, the fusion contrastive module is applied to further strengthen interaction between modalities at molecule level. Finally, we utilize two decoders to perform intra-modal reconstruction.

prehensive understanding of the combined data. The output of the unified encoder is $\mathbf{H}_{fuse}$.

C. Fusion Contrastive Learning(FCL) Module

In this module, we perform cross-modal contrastive to obtain more comprehensive global features by aligning the representations from both modalities at the molecule level. We utilize a READOUT function to obtain the global graph representations for both 2D and 3D modalities.

$$\mathbf{Z}^{2D} = \text{READOUT} \{ \mathbf{h}_{2D}^v \mid v \in \mathcal{V} \} \quad (2)$$

where $\mathbf{h}_{2D}^v$ denote the node embedding. We can also obtain 3D and fused graph representations $\mathbf{Z}^{fuse}$, $\mathbf{Z}^{3D}$, respectively.

We conduct contrastive learning between the fused representation derived from both modalities $\mathbf{Z}^{fuse}$ and the individual modality representations, i.e., $\mathbf{Z}^{2D}$ and $\mathbf{Z}^{3D}$. Taking the topology modality as an example, we denote $\{ \mathbf{Z}_i^{2D}, \mathbf{Z}_i^{fuse} \}$ as positive pairs, and the rest $\{ \mathbf{Z}_i^{2D}, \mathbf{Z}_j^{fuse} \}_{j \neq i}$ as negative pairs. We used the cosine distance to measure the similarity between pairs.

Then the contrastive loss under the topology modality is:

$$\mathcal{L}_{con}^{2D} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp \left(\cos \left(\mathbf{Z}_i^{2D}, \mathbf{Z}_i^{fuse} \right) / \tau \right)}{\sum_{j=1, j \neq i}^N \exp \left(\cos \left(\mathbf{Z}_i^{2D}, \mathbf{Z}_j^{fuse} \right) / \tau \right)}, \quad (3)$$

where τ is a temperature coefficient. Similarly, we can also compute the contrastive loss between geometric modality and fused representation:

$$\mathcal{L}_{con}^{3D} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp \left(\cos \left(\mathbf{Z}_i^{3D}, \mathbf{Z}_i^{fuse} \right) / \tau \right)}{\sum_{j=1, j \neq i}^N \exp \left(\cos \left(\mathbf{Z}_i^{3D}, \mathbf{Z}_j^{fuse} \right) / \tau \right)}, \quad (4)$$

The final contrastive loss is:

$$\mathcal{L}_{con} = \mathcal{L}_{con}^{2D} + \mathcal{L}_{con}^{3D}. \quad (5)$$

By optimizing the alignment between these global molecule-level representations, the model strengthens the interaction between the 2D and 3D features, while preserving the unique characteristics of each modality. This approach complements the fine-grained fusion by ensuring that both atomic-level and molecular-level information are effectively integrated and aligned.

D. Intra-Modal Reconstruction Module

After feature fusion, we split the output features $\mathbf{H}_{fuse}$ into $\mathbf{H}_{2D}$ and $\mathbf{H}_{3D}$. And then, we pad special mask tokens to make up the previously dropped nodes' hidden representations.

2D Decoder. Recent studies [15], [16] have revealed that the decoder architecture and type of reconstruction target strongly impact the encoder's representation learning. Inspired by previous works, we devise a smaller version of the GraphTrans (GTS) architecture as decoder and employ a simplified GNN [17] to obtain the features of reconstruction targets (Simple

GNN Tokenizer), in which the nonlinear update functions are removed. For a molecule, the features of target nodes to be reconstructed are shown as follows:

$$\begin{aligned} \mathbf{H}_i^{(0)} &= \text{Embedding}(\mathbf{x}_i) \in \mathbb{R}^d, \quad \forall i \in \mathcal{V}, \\ \hat{\mathbf{H}}(l) &= \omega(\mathbf{A}) \cdot \mathbf{H}^{(l-1)} \in \mathbb{R}^{|\mathcal{V}| \times d}, \quad 1 \leq l \leq k, \\ \mathbf{H}(l) &= \text{BatchNorm}(\hat{\mathbf{H}}^{(l)}), \end{aligned} \quad (6)$$

$$\text{Target}(g) = \{\mathbf{y}_i = \text{SG}([\mathbf{H}_i^{(1)}, \dots, \mathbf{H}_i^{(k)}]) | i \in \mathcal{V}_m\},$$

where $\mathcal{V}$ denotes the set of nodes and $\mathcal{V}_m$ is the set of masked nodes; $\text{Embedding}(\cdot)$ is a linear layer that uses the weights of the encoder’s node embedding function; $\mathbf{H}_i^{(l)}$ is the i -th row of $\mathbf{H}(l)$; $\text{BatchNorm}(\cdot)$ is a standard Batch Normalization layer without the trainable scaling and shifting parameters [18]; $\omega(\mathbf{A})$ is the original GNN’s aggregation function.

Our loss function, denoted as $\mathcal{L}_{recon}^{2D}$, computes the mean squared error (MSE) between the reconstructed and target features calculated by Equation 6. We compute the loss only on masked tokens.

3D Decoder. The goal of the 3D decoder is to predict the added noise based on the latent embeddings [19], [20]. Let $\mathbf{R} = \{\mathbf{r}_1, \mathbf{r}_2, \dots\}$, $\mathbf{r}_i \in \mathbb{R}^3$ denotes the atom positions of a molecule, and the noisy version of the atom positions is $\mathbf{r}_i + \sigma \epsilon_i$, $i \in \mathbf{R}_m$, where σ is the scaling factor of noise, $\mathbf{R}_m$ denotes the set of corrupted positions. The prediction of the decoder is $\{\hat{\epsilon}_1, \dots, \hat{\epsilon}_m\}$. In this work, we use the Transformer architecture as a decoder. The attention matrix is defined as:

$$\mathbf{A}(\mathbf{X}) = \text{softmax}\left(\frac{\mathbf{X}\mathbf{W}_Q(\mathbf{X}\mathbf{W}_K)^\top}{\sqrt{d}}\right) \quad (7)$$

Following [21], we employ an SE(3) equivariant attention layer as the prediction head:

$$\epsilon_i^k = \left(\sum_{v_j \in V} a_{ij} \Delta_{ij}^k \mathbf{X}_j^{(L)} \mathbf{W}_N^1 \right) \mathbf{W}_N^2, \quad k = 0, 1, 2 \quad (8)$$

where $\mathbf{X}_j^{(L)}$ is the output of the last Transformer block, a_{ij} is the attention score between atom i and atom j calculated by Equation 7, Δ_{ij}^k is the k -th element of the directional vector $\frac{\mathbf{r}_i - \mathbf{r}_j}{\|\mathbf{r}_i - \mathbf{r}_j\|}$ between atom i and atom j , and $\mathbf{W}_N^1 \in \mathbb{R}^{d \times d}$, $\mathbf{W}_N^2 \in \mathbb{R}^{d \times 1}$ are learnable weight matrices. The loss of a batch of molecules $\mathcal{S} = \{V^1, \dots, V^{|\mathcal{S}|}\}$ is the cosine similarity between the predicted noises and the ground-truth noises:

$$\mathcal{L}_{recon}^{3D} = \frac{1}{|\mathcal{S}|} \sum_{V^i \in \mathcal{S}} \sum_{j \in V_m^i} (1 - \cos(\epsilon_j^i, \hat{\epsilon}_j^i)) \quad (9)$$

The overall loss of our model is the sum of the previous loss terms:

$$\mathcal{L}_{total} = \alpha_1 \cdot \mathcal{L}_{con} + \alpha_2 \cdot \mathcal{L}_{recon}^{2D} + \alpha_3 \cdot \mathcal{L}_{recon}^{3D} \quad (10)$$

where $\alpha_1, \alpha_2, \alpha_3$ are weighting coefficients. With such design, our model can learn 3D and 2D features independently while preserving strong multiscale (atomic-level and molecule-level) interactions between the two modalities.

III. EXPERIMENTS

A. Experimental Settings

Datasets. We pretrain our model on the PCQM4Mv2 training set from OGB Large-Scale Challenge. It is a sub-dataset of PubChemQC, which contains 3.4 million molecules with both 2D topological graphs and 3D geometric conformations. With the pre-trained model, we conduct experiments on 23 molecular tasks with different data formats and evaluate its versatility and effectiveness. We study two representative tasks: MoleculeNet (2D, 8 classification tasks and 3 regression tasks), QM9 quantum properties (3D, 12 tasks).

B. Molecular Property Prediction Task (2D)

We assess our model using MoleculeNet, a widely used benchmark for 2D molecular property prediction, encompassing a variety of molecular properties, from quantum mechanics and physical chemistry to biophysics and physiology. The evaluation follows the scaffold split method [22] and we report the mean and standard deviation of results based on three different random seeds.

Table I represents the performance of our method across 8 biological classification tasks in MoleculeNet. Impressively, our approach attains state-of-the-art performance in 6 out of the 8 tasks, with large margins in certain cases (e.g., 85.4 v.s. 83.7 on Bace). Furthermore, Table II demonstrates that our method achieves the best performance across all 3 physicochemical regression tasks in MoleculeNet. Compared with other multi-modal methods (3D infoMax [9], GraphMVP [7], MoleculeSDE [8]), which utilize contrastive learning as one of the objectives to align molecules at the molecule level, the improvement observed in our method can be attributed to the atomic interaction between two modalities (2D topology and 3D geometry). Moreover, our method also surpasses all 2D baselines (as shown in the upper section of the table), highlighting that the integration of 3D information enhances molecular properties prediction.

C. Quantum Property Prediction Task (3D)

We utilize the QM9 dataset [23] to evaluate our model on molecular tasks with the 3D data format. QM9 is a dataset with 134K molecules consisting of 9 heavy atoms. It has 12 tasks related to quantum properties, including the energetic, electronic and thermodynamic properties of molecules. Following [24], we randomly select 10,000 molecules for validation and 10,831 molecules for testing. The remaining molecules are used to fine-tune our model. As shown in Table III, we present experimental results. Our model achieves state-of-the-art performance on 9 out of 12 tasks, with substantial margins in certain cases (e.g., G298, U298, U0), showcasing our model’s robust capability for 3D tasks.

D. Ablation Studies

Model architecture We conduct an ablation study on key modules of MolInterAct, including the Fine-grained Cross-modal Fusion (FCF) module and the Fusion Contrastive Learning (FCL) module. Removing the FCF module reduces the

TABLE I: Results of 2D molecular property prediction tasks on MoleculeNet. The evaluation is Mean (and Standard Deviation) of the ROC-AUC score (higher is better) under scaffold splitting.

Pretraining	BBBP $\uparrow$	Tox21 $\uparrow$	ToxCast $\uparrow$	SIDER $\uparrow$	ClinTox $\uparrow$	MUV $\uparrow$	HIV $\uparrow$	Bace $\uparrow$	Avg $\uparrow$
AttrMask (2020)	65.0 $\pm$ 2.3	74.8 $\pm$ 0.2	62.9 $\pm$ 0.1	61.2 $\pm$ 0.1	87.7$\pm$1.1	73.4 $\pm$ 2.0	76.8 $\pm$ 0.5	79.7 $\pm$ 0.3	72.7
ContextPred (2020)	65.7 $\pm$ 0.6	74.2 $\pm$ 0.0	62.5 $\pm$ 0.3	62.2 $\pm$ 0.5	77.2 $\pm$ 0.8	75.3 $\pm$ 1.5	77.1 $\pm$ 0.8	76.0 $\pm$ 2.0	71.3
GraphCL (2020)	69.7 $\pm$ 0.6	73.9 $\pm$ 0.6	62.4 $\pm$ 0.5	60.5 $\pm$ 0.8	76.0 $\pm$ 2.6	69.8 $\pm$ 2.6	78.5 $\pm$ 1.2	75.4 $\pm$ 1.4	70.8
InfoGraph(2020)	67.5 $\pm$ 0.1	73.2 $\pm$ 0.4	63.7 $\pm$ 0.5	59.9 $\pm$ 0.3	76.5 $\pm$ 1.0	74.1 $\pm$ 0.7	75.1 $\pm$ 0.9	77.8 $\pm$ 0.8	71.0
GROVER (2020)	70.0 $\pm$ 0.1	74.3 $\pm$ 0.1	65.4 $\pm$ 0.4	64.8 $\pm$ 0.6	81.2 $\pm$ 3.0	67.3 $\pm$ 1.8	62.5 $\pm$ 0.9	82.6 $\pm$ 0.7	71.0
MolCLR (2022)	66.6 $\pm$ 1.8	73.0 $\pm$ 0.1	62.9 $\pm$ 0.3	57.5 $\pm$ 1.7	86.1 $\pm$ 0.9	72.5 $\pm$ 2.3	76.2 $\pm$ 1.5	71.5 $\pm$ 3.1	70.8
GraphMAE (2022)	72.0 $\pm$ 0.6	75.5 $\pm$ 0.6	64.1 $\pm$ 0.3	60.3 $\pm$ 1.1	82.3 $\pm$ 1.2	76.3 $\pm$ 2.4	77.2 $\pm$ 1.0	83.1 $\pm$ 0.9	73.9
Mole-BERT (2023)	71.9 $\pm$ 1.6	76.8 $\pm$ 0.5	64.3 $\pm$ 0.2	62.8 $\pm$ 1.1	78.9 $\pm$ 3.0	78.6 $\pm$ 1.8	78.2 $\pm$ 0.8	80.8 $\pm$ 1.4	74.0
3D InfoMax (2022)	69.1 $\pm$ 1.0	74.5 $\pm$ 0.7	64.4 $\pm$ 0.8	60.6 $\pm$ 0.7	79.9 $\pm$ 3.4	74.4 $\pm$ 2.4	76.1 $\pm$ 1.3	79.7 $\pm$ 1.5	72.3
GraphMVP (2022)	68.5 $\pm$ 0.2	74.5 $\pm$ 0.4	62.7 $\pm$ 0.1	62.3 $\pm$ 1.6	79.0 $\pm$ 2.5	75.0 $\pm$ 1.4	74.8 $\pm$ 1.4	76.8 $\pm$ 1.1	71.7
MoleculeSDE (2023)	71.8 $\pm$ 0.7	76.8 $\pm$ 0.3	65.0 $\pm$ 0.2	60.8 $\pm$ 0.3	87.0 $\pm$ 0.5	80.9$\pm$0.3	78.8 $\pm$ 0.9	79.5 $\pm$ 2.1	75.1
MoleBlend (2024)	<u>73.0$\pm$0.8</u>	<u>77.8$\pm$0.8</u>	<u>66.1$\pm$0.0</u>	<u>64.9$\pm$0.3</u>	<u>87.6$\pm$0.7</u>	<u>77.2$\pm$2.3</u>	<u>79.0$\pm$0.8</u>	<u>83.7$\pm$1.4</u>	<u>76.2</u>
MolInterAct (ours)	73.2$\pm$0.3	78.5$\pm$0.4	67.4$\pm$0.2	65.0$\pm$0.6	87.4 $\pm$ 1.0	<u>79.5$\pm$0.3</u>	80.2$\pm$0.2	85.4$\pm$0.2	77.1

The best and second best results are marked by **bold** and underlined.

TABLE II: Results of 2D molecular property prediction regression tasks on MoleculeNet. The evaluation is Mean (and Standard Deviation) of RMSE (lower is better) under scaffold splitting.

Pretraining	ESOL $\downarrow$	FreeSolv $\downarrow$	Lipo $\downarrow$
AttrMask (2020)	1.112 $\pm$ 0.048	-	0.730 $\pm$ 0.004
ContextPred (2020)	1.196 $\pm$ 0.037	-	0.702 $\pm$ 0.020
GROVER (2020)	0.983 $\pm$ 0.090	2.176 $\pm$ 0.052	0.817 $\pm$ 0.008
MolCLR (2022)	1.271 $\pm$ 0.040	2.594 $\pm$ 0.249	0.691 $\pm$ 0.004
3D InfoMax (2022)	0.894 $\pm$ 0.028	2.337 $\pm$ 0.227	0.695 $\pm$ 0.012
GraphMVP (2022)	1.029 $\pm$ 0.033	-	0.695 $\pm$ 0.012
MOLEBLEND (2024)	<u>0.831$\pm$0.026</u>	<u>1.910$\pm$0.163</u>	<u>0.638$\pm$0.004</u>
MolInterAct (ours)	0.821$\pm$0.012	1.851$\pm$0.094	0.635$\pm$0.002

The best and second best results are marked by **bold** and underlined.

model to independently predicting masked tokens within each modality, effectively resulting in a single-modality setup. Thus, for this ablation condition, we use only single-modality input (either 2D or 3D) to examine the role of FCL in fine-grained modality fusion. As shown in Table IV, the model leveraging multiple modalities significantly outperforms single-modality approaches, demonstrating that the FCL module effectively helps the model learn more comprehensive and complementary cross-modal information. Furthermore, removing the FCL module—which aligns global molecular representations—leads to a noticeable performance drop, indicating that this module complements fine-grained representation fusion and collectively enhances the overall model performance.

Complementary masking strategy. To investigate how masking strategies influence cross-modal information interaction during the fusion stage, we compare complementary masking with two conventional strategies: random masking and uniform masking. Random masking independently samples masked regions for 2D and 3D modalities, while uniform masking applies identical masking patterns to both modalities, aligning the masked regions. In contrast, complementary masking adopts an opposite approach by masking non-

overlapping parts across modalities. We implement these three strategies during pretraining and evaluate their performance after fine-tuning on MoleculeNet and QM9. As shown in Table V, complementary masking achieves the best results across all tasks, indicating its effectiveness in enhancing molecular representation learning. Compared to random masking, complementary masking facilitates more meaningful cross-modal interactions between atoms with diverse semantic information, thereby aiding knowledge transfer across modalities. In contrast, uniform masking results in semantically aligned point cloud and topology features, which limits the model’s ability to effectively leverage complementary information from the other modality.

IV. CONCLUSION

We propose MolInterAct, a self-supervised pretraining framework that fosters 2D and 3D interactions at both the atomic and molecular levels. Our method achieves deep and granular interaction between modalities through the fine-grained cross-modal fusion (FCF) module, which operates at the atomic level. Additionally, the fusion contrastive learning (FCL) module enforces molecule-level alignment, capturing the correlations and complementarity between global modality features. These two modules complement each other, resulting in a more robust molecular representation. Our method achieves state-of-the-art performance across a broad range of 2D and 3D benchmarks, demonstrating its effective and highly multiscale interactive multi-modal learning pipeline, which excels in feature extraction for both geometry and topology.

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TABLE III: Results of quantum property prediction tasks on QM9. The evaluation is Mean Absolute Error (MAE, lower is better).

Pretraining	Alpha↓	Gap↓	HOMO↓	LUMO↓	Mu↓	Cv↓	G298↓	H298↓	R2↓	U298↓	U0↓	Zpve↓
Distance Prediction (2022)	0.065	45.9	27.6	23.3	0.031	0.033	14.83	15.81	0.248	15.07	15.01	1.837
3D InfoMax (2022)	0.057	42.1	25.9	21.6	0.028	0.030	13.73	13.62	0.141	13.81	13.30	1.670
GraphMVP (2022)	0.056	42.0	25.8	21.6	0.027	0.029	13.43	13.31	0.136	13.03	13.07	1.609
MoleculeSDE (2023)	0.054	41.8	25.7	21.4	0.026	0.028	13.07	12.05	0.151	12.54	12.04	1.587
MOLEBLEND (2024)	0.060	34.8	<u>21.5</u>	19.2	0.037	0.031	12.44	11.97	0.417	12.02	11.82	1.580
SE(3)-DDM (2023)	<u>0.046</u>	40.2	23.5	19.4	<u>0.015</u>	<u>0.024</u>	<u>7.65</u>	<u>7.09</u>	0.122	<u>6.99</u>	<u>6.92</u>	<u>1.31</u>
3D-EMGP (2023)	0.057	37.1	21.3	<u>18.2</u>	0.020	0.026	9.30	8.70	0.092	8.60	8.60	1.38
MolInterAct (ours)	0.043	34.3	23.6	17.1	0.010	0.022	6.85	6.10	<u>0.120</u>	5.60	5.51	1.24

The best and second best results are marked by **bold** and underlined.

TABLE IV: Comparison of multi-modal and single-modal (ROC-AUC, ↑ and MAE, ↓) on MoleculeNet and QM9.

Input	MoleculeNet			QM9		
	ToxCast↑	SIDER↑	Bace↑	Alpha↓	G298↓	Zpve↓
2D	65.5	62.8	82.7	-	-	-
3D	-	-	-	0.051	7.63	1.54
2D+3D w/o FCL	66.8	63.8	84.0	0.046	7.41	1.35
2D+3D	67.4	65.0	85.4	0.043	6.85	1.24

TABLE V: Comparison of cross-modal masking strategies (ROC-AUC, ↑ and MAE, ↓) on MoleculeNet and QM9.

Masking Strategy	MoleculeNet			QM9		
	ToxCast↑	SIDER↑	Bace↑	Alpha↓	G298↓	Zpve↓
Random	65.9	63.4	83.4	0.046	7.60	1.35
Uniform	66.8	63.8	82.9	0.047	7.20	1.28
Complementary	67.4	65.0	85.4	0.043	6.85	1.24

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